

Brent Crude Mean-Reversion Under OPEC+ Band Enforcement

Stochastic Differential Equation — Derivation and Simulation Framework

Geeconomic SDE Project · Case 1

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1 Observation and Motivation

Brent crude does not random-walk indefinitely. Over medium horizons (weeks to months) it exhibits persistent mean-reversion toward a structural band of approximately \$70–\$90. This reversion is not statistical — it is enforced by real production decisions made by two actors with identifiable and stable incentive structures.

The question this model answers is:

Given Brent currently at \$82, what is the full probability distribution of price at 30, 60, and 90 days from now, under the structural constraints imposed by OPEC+ and US shale?

This is not a point forecast. The output is a probability distribution over price levels at each horizon — directly tradeable against market option pricing.

2 Economic Hypothesis

2.1 The Two-Actor Band Mechanism

Floor (\$70–75): OPEC+ member fiscal breakevens cluster near this range (Saudi Arabia ≈\$70, Iraq ≈\$75). Below it, members run budget deficits and have a strong collective incentive to cut production. The cut is a *rate* decision — supply is reduced continuously until price recovers. Crucially, the response is proportional: a small breach triggers modest cuts; a large breach triggers coordinated emergency response (as documented in 2020 and 2022 cut cycles).

Ceiling (\$85–90): Above this range, US shale operators can profitably hedge forward production. Rig counts rise and new supply enters within 3–6 months, continuously capping upside. Again the response is proportional — higher prices bring more rigs online, not a binary switch.

The proportionality of both responses is what makes price *mean-reverting* rather than merely bounded. There is no wall that price bounces off — there is a restoring force that grows stronger the further price moves from the band center.

2.2 Why Volatility Is State-Dependent

Market uncertainty is not uniform across the price band. Near \$70 and \$90, OPEC meetings, shale hedging reports, and inventory data are watched closely — implied volatility is elevated,

realized vol is high. In the middle of the band (\$76–\$84), markets are calm and vol is low. Therefore σ must be a function of distance from θ , not a constant. This is the key nonlinearity and the primary source of analytical intractability.

2.3 Jump Sources

Geopolitical events arrive discretely, not continuously: surprise OPEC+ production decisions, tanker route disruptions (e.g. Houthi attacks), sudden China demand revisions from PMI data. A continuous diffusion term alone cannot replicate these — an explicit jump component is required.

2.4 State Variable Choice

Let $X(t) = \log(\text{Brent spot price at time } t)$. We work in log-price rather than price for two reasons: (i) price cannot go negative, and (ii) percentage moves are the economically meaningful unit — a \$5 move at \$40 is very different from \$5 at \$80.

3 SDE Derivation

3.1 Building the Expression Term by Term

Step 1 — Drift. The restoring force argument says: the rate of change of X is pulled toward θ at a speed proportional to how far it currently is from θ .

“If price is above target, drift is negative. If below, drift is positive. The further away, the stronger the pull.”

$$\mu(X, t) = \kappa(\theta - X) \quad (1)$$

This is the Ornstein-Uhlenbeck drift. κ controls reversion speed. A value of $\kappa = 2.1 \text{ yr}^{-1}$ implies a half-life of $\ln(2)/\kappa \approx 4$ months, consistent with observed OPEC response cycles.

Step 2 — Diffusion. Constant σ would assert that uncertainty is uniform across all price levels. Our mechanism says otherwise. We encode this as:

$$\sigma(X, t) = \sigma_0 \cdot (1 + \alpha |X - \theta|) \quad (2)$$

where σ_0 is baseline annualised volatility and α controls how quickly vol scales with distance from the mean. Minimum uncertainty is at $X = \theta$; it rises as price deviates in either direction.

Step 3 — Jump. Geopolitical shocks arrive as a Poisson process with intensity λ (average events per year). Each shock moves log-price by $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$. This is a compound Poisson process:

$$dq_t : \quad P(dq = 1) = \lambda dt, \quad J \sim \mathcal{N}(\mu_j, \sigma_j^2) \quad (3)$$

3.2 Full SDE

Assembling the three components:

$$\boxed{dX = \kappa(\theta - X) dt + \sigma_0(1 + \alpha|X - \theta|) dW + J dq} \quad (4)$$

Term	Type	Economic meaning
$\kappa(\theta - X) dt$	Drift	OPEC/shale restoring force
$\sigma_0(1 + \alpha X - \theta) dW$	Diffusion	State-dependent market uncertainty
$J dq$	Jump	Discrete geopolitical shock

4 Analytical Tractability Check

Before building a numerical solver, we attempt the analytical path and identify exactly where it fails. This is the most important intellectual step — it justifies the numerical approach rigorously.

Q1: Is the diffusion coefficient constant?

No. $\sigma(X) = \sigma_0(1 + \alpha|X - \theta|)$ is nonlinear in X . The term $|X - \theta|$ has derivative $\text{sign}(X - \theta)$ — a step function, discontinuous at $X = \theta$.

Applying Itô's lemma to any transformation $f(X)$ requires computing:

$$df = f'(X) dX + \frac{1}{2} f''(X) (dX)^2 \quad (5)$$

The quadratic variation term produces $\frac{1}{2} f''(X) \cdot \sigma(X)^2 dt$. With $\sigma(X)$ nonlinear, $\sigma^2 = \sigma_0^2(1 + \alpha|X - \theta|)^2$ is a nonlinear function of a stochastic process. The resulting Fokker-Planck PDE:

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x} [\kappa(\theta - x) p] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma_0^2(1 + \alpha|x - \theta|)^2 p] \quad (6)$$

has a non-smooth diffusion coefficient and admits no closed-form transition density.

→ **FAILS at Q1. Analytical solution does not exist.**

Q2: Is drift linear in X ? Yes — $\kappa(\theta - X)$ is linear. This alone (classical OU) would permit an analytic solution, but Q1 already failed.

Q3: Are jumps tractable in isolation? λ is constant and $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$, so Merton's formula would apply to the jump component alone. Irrelevant here — Q1 already forces numerical treatment.

Decision: Euler-Maruyama numerical solver required.

5 Numerical Solution: Euler-Maruyama

5.1 Discretisation

Over time step $\Delta t = 1/252$ (one trading day):

$$X(t + \Delta t) = X(t) + \kappa(\theta - X(t)) \cdot \Delta t + \sigma_0(1 + \alpha|X(t) - \theta|) \cdot \sqrt{\Delta t} \cdot Z + J \cdot \mathbf{1}_{[\text{Poisson}(\lambda \Delta t) \geq 1]} \quad (7)$$

where $Z \sim \mathcal{N}(0, 1)$ and $J \sim \mathcal{N}(\mu_j, \sigma_j^2)$. This is the exact update rule implemented in `core/euler_maruyama.py`.

5.2 Implementation Architecture

The solver follows a model-injection pattern. The core loop in `euler_maruyama.py` calls three functions at each step:

Function	Implements
<code>model.drift(X, t)</code>	$\kappa(\theta - X)$
<code>model.diffusion(X, t)</code>	$\sigma_0(1 + \alpha X - \theta)$
<code>model.jump(X, t, rng, dt)</code>	Compound Poisson draw

Each case only overrides these three functions. The solver loop is shared across all five cases in the project.

6 Parameters

Parameter	Value	Units	Reasoning
κ	2.1	yr ⁻¹	≈4 month half-life, matches OPEC response cycle
θ	log(78)	—	Midpoint of OPEC+ fiscal breakeven band
σ_0	0.18	—	Baseline annualised vol, historical Brent
α	0.40	—	Vol scaling near band boundaries
λ	3.0	yr ⁻¹	≈3 material geopolitical shocks per year
μ_j	0.0	—	Shocks symmetric on average
σ_j	0.06	—	Typical ±6% log-price move per shock event

Parameters are currently set by economic reasoning. Formal MLE calibration to historical Brent data is to be implemented in `calibration/estimate.py`.

7 Results and Inference

7.1 Simulation Setup

$X_0 = \log(82)$, horizon $T = 0.5$ years, $\Delta t = 1/252$, $N = 10,000$ Monte Carlo paths.

7.2 Scenario Probability Table

Horizon	P(<\$70)	P(>\$90)	P(in band)	Median	5th pct	95th pct
30d	1.3%	5.4%	93.4%	\$81.0	\$73.2	\$90.3
60d	4.7%	8.7%	86.7%	\$80.3	\$70.3	\$92.1
90d	7.0%	9.3%	83.7%	\$79.7	\$68.8	\$93.1

7.3 Reading the Output

Median path. Moves $\$82 \rightarrow \$81.0 \rightarrow \$80.3 \rightarrow \79.7 , drifting toward $\theta = \$78$. Mean-reversion is working as specified: starting above target, the drift term $\kappa(\theta - X)$ is negative at every step.

Asymmetry. At 30 days, $P(>\$90) = 5.4\%$ versus $P(<\$70) = 1.3\%$. Starting at $\$82$, the ceiling is geometrically closer than the floor in log-price terms — less movement required to breach $\$90$ than $\$70$. This asymmetry is structural, not assumed.

Tail widening. The band probability decays from 93.4% to 83.7% over 90 days as jump events and state-dependent diffusion accumulate tail paths over time.

7.4 Tradeable Inference

Standard Black-Scholes options pricing uses a single implied volatility number and a log-normal assumption. It has no concept of structural levels at \$70 and \$90 and no mean-reversion term. It therefore prices equidistant strikes approximately symmetrically.

Our model says the ceiling breach (5.4%) is $\approx 4\times$ more likely than the floor breach (1.3%) at 30 days, purely from starting position and structural geometry. If the options market is pricing \$90 calls and \$70 puts at similar implied volatilities — treating both tails as equally probable — then \$90 calls are underpriced relative to our model.

The trade: long \$90 calls, short \$70 puts. The thesis has a timing constraint — by 90 days the two tail probabilities converge (7.0% vs 9.3%) and the skew argument weakens. The model specifies both the direction and the horizon of the edge.

8 Simulation Output

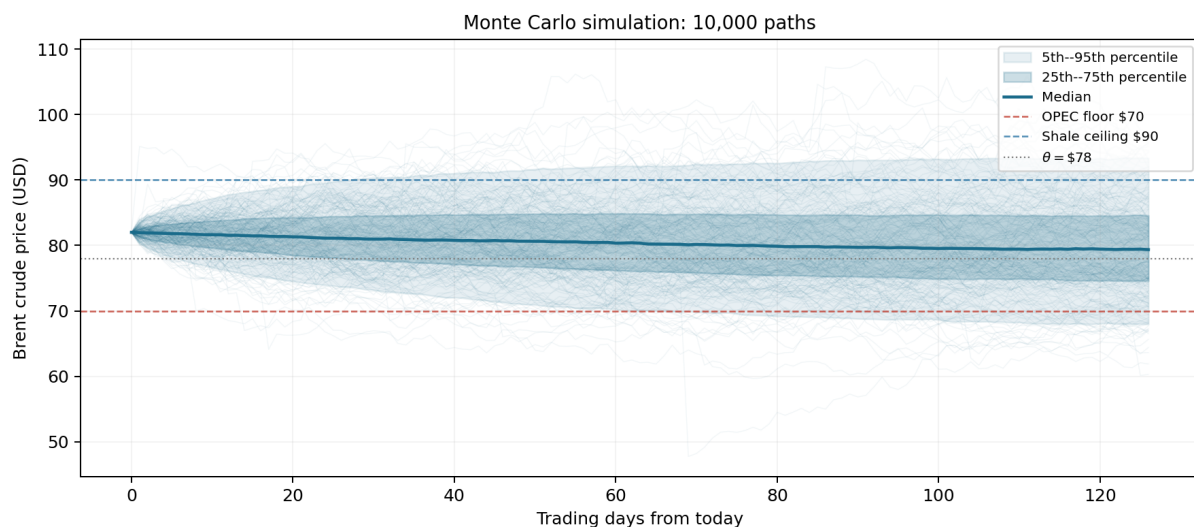


Figure 1: 10,000 Monte Carlo paths for Brent crude, converted to price space. Shaded bands show the 5th–95th and 25th–75th percentile ranges. The median path drifts from \$82 toward $\theta = \$78$ as mean-reversion takes effect. Dashed lines mark the OPEC floor (\$70) and shale ceiling (\$90). Occasional sharp single-step moves in individual paths are jump events.

9 Learning, Application, and Edge

9.1 What This Case Teaches

Classical OU has constant σ — it has a closed-form solution and a numerical solver is never needed. The moment σ becomes a function of X , the Fokker-Planck PDE loses its Gaussian structure and analytical tractability breaks. This case is deliberately constructed around that boundary. Building Euler-Maruyama from scratch here — rather than calling a library — means understanding exactly what the solver is doing at each step and why it exists. That understanding transfers directly to every subsequent case where the equations are more complex.

9.2 Where It Applies

1. An oil-importing sovereign treasury sizing a 90-day hedge program needs tail probabilities at specific price levels, not a point forecast.
2. A commodity desk running scenario P&L ahead of an OPEC meeting needs a distribution over outcomes, not a single expected value.
3. A macro fund with a long energy position needs the probability of breaching a stop level within a given horizon under current geopolitical conditions.

9.3 Edge Over Standard Models

Three specific departures from Black-Scholes, each producing a measurable difference in output:

State-dependent volatility. σ rises as price approaches band boundaries. Black-Scholes spreads uncertainty uniformly across all price levels. This model concentrates it where market stress actually occurs, producing heavier tails at structurally meaningful levels.

Structural band levels. The drift term has explicit knowledge of \$70 and \$90 as economically enforced boundaries. Black-Scholes has no concept of structural levels — it treats all price regions identically.

Jump tail mass. Discrete geopolitical shocks place probability mass in the tails as sudden discontinuous moves. Log-normal diffusion spreads that mass smoothly across the distribution, systematically underpricing sharp event risk.

10 Project Context

This is Case 1 of a five-case Geoeconomic SDE framework. Each case introduces one new numerical technique on top of the last:

Case	Asset / Scenario	New technique introduced
1	Brent crude under OPEC+	Euler-Maruyama, state-dependent diffusion
2	EM FX under sanctions shock	Poisson jump process, compound arrivals
3	Yield curve $\leftrightarrow$ equity	Cholesky decomposition, correlated SDEs
4	Commodity regime switching	Markov chain, conditional parameter switching
5	Geoeconomic scenario engine	Full 4D Monte Carlo, scenario scoring

Repository: `geoeconomic-sde/cases/01_oil_opec/` · *Core solver:* `core/euler_maruyama.py` · *Parameters:* `calibration/params.json`